Multi-Class Text Classification: A Comparative Analysis

Group 3

***Abstract*—This project investigates multi-class news headline classification using machine learning, and deep learning models. Three preprocessing strategies were tested: no preprocessing, extreme preprocessing, and optimum preprocessing. TF-IDF and Word2Vec Skip-gram embeddings were used for text representation. Eight models were evaluated: Logistic Regression, DNN, SimpleRNN, GRU, LSTM, Bidirectional SimpleRNN, Bidirectional GRU, and Bidirectional LSTM. Results show that TF-IDF with Logistic Regression achieved the best test accuracy of 93.28% and macro F1-score of 93.24%, outperforming all deep learning models. Optimum preprocessing produced the best overall result because it removed noise without destroying important topic-specific features.**

***Index Terms*—NLP, Text Classification, TF-IDF, Word2Vec, Logistic Regression, RNN, GRU, LSTM**

I. Introduction

Automatic news categorization is an important NLP task used in news aggregation, recommendation systems, and information retrieval. Since large volumes of news are published daily, manual classification is inefficient. This project focuses on classifying short news headlines into predefined topic categories.

News headlines are short but information-rich, making classification challenging due to limited context, vocabulary variation, and overlapping topic-specific terms. This project compares classical machine learning and deep learning models under different preprocessing settings to identify the best combination of preprocessing, representation, and model architecture.

II. Methodology

A. Dataset and EDA

The dataset contains labeled news headlines with two main columns: news headline and news topic. It is divided into training and test sets. The training set contains four topic classes: Science and Technology, Business, Sports, and World News. The test set is balanced with 3000 samples per class.

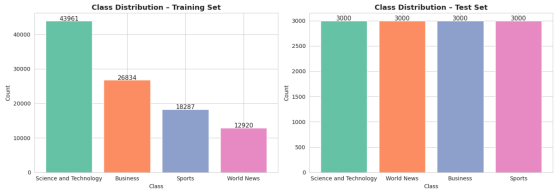


Fig. 1: Class distribution of training and test sets

EDA showed that the training data is imbalanced, with Science and Technology having the highest number of samples and World News having the lowest. However, the test set is balanced. Headline length analysis showed that most headlines contain a moderate number of words, usually concentrated around short text lengths.

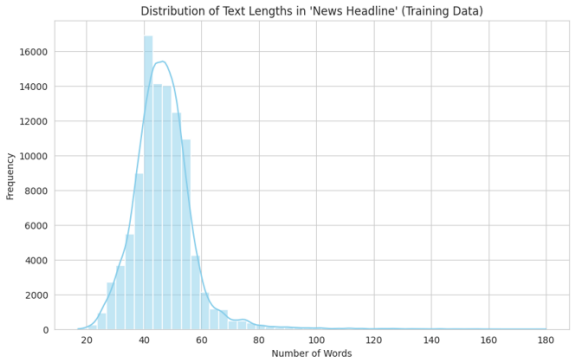


Fig. 2: Distribution of headline lengths

The word cloud and n-gram analysis revealed many noisy HTML-related tokens such as html, body, and br. These findings motivated the need for careful preprocessing.

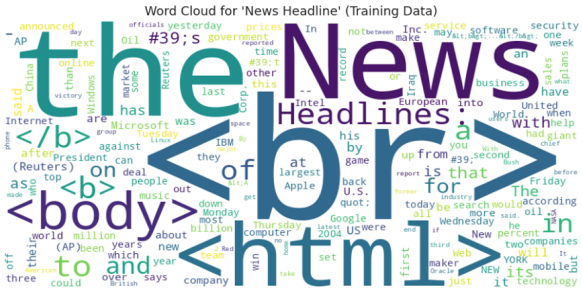


Fig. 3: Word cloud of training headlines

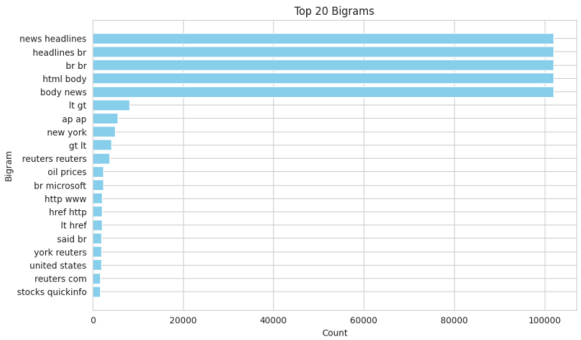


Fig. 4: Top 20 bigrams

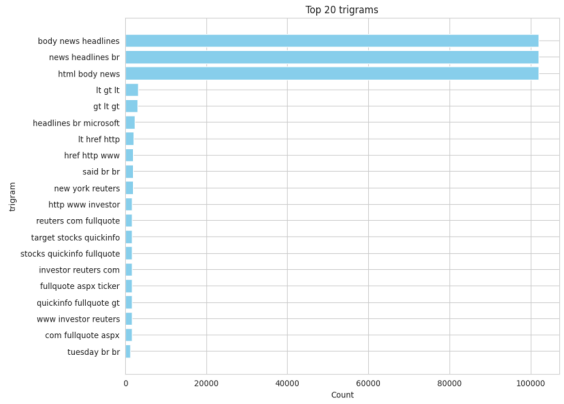


Fig. 5: Top 20 trigrams

B. Preprocessing

Three preprocessing strategies were applied. First, no preprocessing used the raw headline text. Second, extreme preprocessing included lowercasing, HTML and URL removal, punctuation removal, digit removal, stopword removal, short-token filtering, and stemming. Third, optimum preprocessing used balanced cleaning: lowercasing, HTML and URL removal, special character removal, tokenization, selective stopword removal, and removal of domain noise tokens such as "news" and "headlines". Stemming was not used in optimum preprocessing because it can remove useful morphological information from short texts.

C. Text Representation

Two word representation methods were used. TF-IDF represented headlines as sparse vectors using a maximum of 20,000 features, unigram and bigram range, and minimum document frequency of 2. Word2Vec Skip-gram was used to create 100-dimensional dense embeddings with window size 5 and minimum count 2. These embeddings were used in the recurrent neural network models.

D. Models

Eight models were trained. TF-IDF features were used with Logistic Regression and DNN. The DNN architecture included dense layers of 256, 128, 64, and 32 neurons with ReLU activation and dropout. Skip-gram embeddings were used with SimpleRNN, GRU, LSTM, Bidirectional SimpleRNN, Bidirectional GRU, and Bidirectional LSTM. All sequence models used maximum vocabulary size 10,000 and maximum sequence length 100. Adam optimizer, sparse categorical cross-entropy loss, and early stopping were applied.

III. Results

This section presents the experimental results and comparative analysis of all models. The analysis focuses on preprocessing impact, overall model performance, classical machine learning versus deep learning, recurrent model comparison, training behavior, and confusion matrix observations.

A. Effect of Preprocessing

Table I shows the performance of TF-IDF with Logistic Regression under three preprocessing settings.

TABLE I: Effect of Preprocessing on TF-IDF + Logistic Regression

Preprocessing	Accuracy	Precision	Recall	F1
No Preprocessing	0.9215	0.9216	0.9215	0.9210
Extreme	0.9037	0.9041	0.9037	0.9031
Optimum	**0.9328**	**0.9329**	**0.9328**	**0.9324**

The results show that optimum preprocessing achieved the best performance, with 93.28% accuracy and 93.24% F1-score. It improved accuracy by 1.13 percentage points compared with no preprocessing and by 2.91 percentage points compared with extreme preprocessing. This indicates that balanced preprocessing is more suitable for short news headlines.

Extreme preprocessing produced the lowest result because aggressive cleaning removed useful information. Stemming can merge different meaningful words into the same root form, while digit removal can remove important signals from sports scores or business headlines. No preprocessing still performed reasonably well because raw headlines already contain strong topic-specific words, but it also keeps unnecessary noise. Therefore, optimum preprocessing gives the best balance between noise removal and information preservation.

B. Overall Model Comparison

Table II shows the final test performance of all models using optimum preprocessing.

TABLE II: Test Performance of All Models

Model	Acc.	Prec.	Rec.	F1
TF-IDF + LR	**0.9328**	**0.9329**	**0.9328**	**0.9324**
TF-IDF + DNN	0.9221	0.9222	0.9221	0.9214
Skip-gram + SimpleRNN	0.8803	0.8798	0.8803	0.8794
Skip-gram + GRU	0.9135	0.9156	0.9135	0.9134
Skip-gram + LSTM	0.9135	0.9156	0.9135	0.9134
Skip-gram + Bi-RNN	0.8737	0.8806	0.8737	0.8739
Skip-gram + Bi-GRU	0.9115	0.9140	0.9115	0.9114
Skip-gram + Bi-LSTM	0.9171	0.9207	0.9171	0.9171

TF-IDF with Logistic Regression achieved the highest test accuracy of 93.28%, making it the best-performing model in this experiment. TF-IDF with DNN achieved 92.21% accuracy and was the second-best model. Although DNN is more complex, it did not outperform Logistic Regression, showing that higher model complexity does not always improve performance when the feature representation is already strong.

Among the recurrent models, Bi-LSTM achieved the best result with 91.71% accuracy. GRU and LSTM both achieved 91.35% accuracy, showing that both gated architectures performed similarly. Bi-GRU achieved 91.15%, slightly lower than standard GRU. SimpleRNN and Bi-SimpleRNN produced the weakest results, with 88.03% and 87.37% accuracy respectively.

The accuracy comparison shows a clear ranking among model groups. TF-IDF-based models performed better than

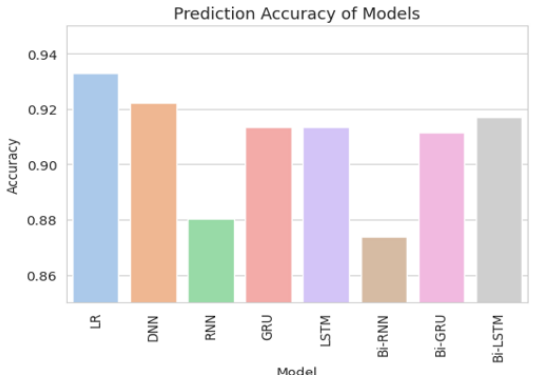


Fig. 6: Prediction accuracy comparison of models

Skip-gram-based recurrent models. This suggests that representation quality played a larger role than model complexity. Since news headlines are short, topic-specific keywords and bigrams are very important. TF-IDF captures these signals directly, while Word2Vec embeddings trained only on the project dataset may not provide enough semantic richness.

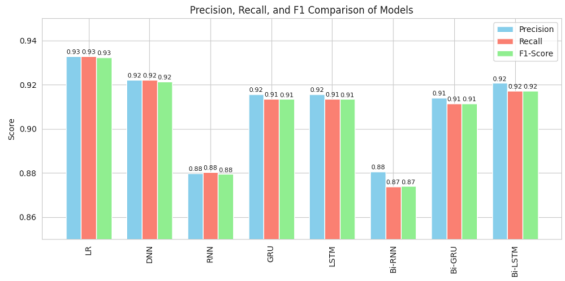


Fig. 7: Precision, recall, and F1-score comparison

The precision, recall, and F1-score comparison shows that Logistic Regression produced the most balanced performance. Its precision, recall, and F1-score are almost equal, which means the model did not strongly over-predict or under-predict any class overall. DNN also performed strongly but remained slightly below Logistic Regression. The recurrent models showed more variation, especially SimpleRNN and Bi-SimpleRNN, which had lower F1-scores.

C. Model Ranking and Interpretation

TABLE III: Ranked Model Comparison

Rank	Model	Accuracy	F1-Score
1	TF-IDF + Logistic Regression	**0.9328**	**0.9324**
2	TF-IDF + DNN	0.9221	0.9214
3	Skip-gram + Bi-LSTM	0.9171	0.9171
4	Skip-gram + GRU	0.9135	0.9134
5	Skip-gram + LSTM	0.9135	0.9134
6	Skip-gram + Bi-GRU	0.9115	0.9114
7	Skip-gram + SimpleRNN	0.8803	0.8794
8	Skip-gram + Bi-SimpleRNN	0.8737	0.8739

The ranking confirms that the best model is TF-IDF with Logistic Regression. This model is simple, efficient, and highly accurate. Its strong performance is mainly due to the nature of the task. News headlines usually contain a small number of highly informative words. For example, sports headlines often contain words related to teams, matches, goals, or players, while business headlines often contain financial and market-related terms. TF-IDF assigns higher importance to such discriminative words and phrases.

The DNN model used the same TF-IDF representation but performed slightly worse. This suggests that the decision boundary in TF-IDF space is mostly linear, so Logistic Regression is already sufficient. The extra hidden layers of DNN may increase complexity without providing a meaningful advantage.

The recurrent models used Skip-gram Word2Vec embeddings. These models can learn word order and sequence information, but they performed lower than TF-IDF-based models. One possible reason is that the embeddings were trained only on the available dataset, so they may not capture broad semantic relationships as well as large pretrained embeddings. Also, short headlines may not require complex sequential modeling as much as longer documents.

D. Classical Machine Learning vs. Deep Learning

The most important finding is that the classical machine learning model outperformed all deep learning models. TF-IDF with Logistic Regression achieved 93.28% accuracy, while the best deep learning model, Bi-LSTM, achieved 91.71%. This creates a 1.57 percentage point gap in favor of Logistic Regression.

This result is meaningful because it shows that deep learning is not always the best choice for every NLP task. For short-text classification, sparse keyword-based features can be extremely effective. TF-IDF captures the importance of words and bigrams across the corpus, and Logistic Regression can classify them efficiently. In contrast, recurrent models process text sequentially and depend heavily on the quality of word embeddings.

The comparison also shows that TF-IDF with DNN was strong but still lower than Logistic Regression. This means that the advantage came mainly from the TF-IDF representation, not from deep model complexity. Therefore, for this dataset, feature representation had more impact than architecture depth.

E. Recurrent Architecture Comparison

TABLE IV: Recurrent Model Comparison

Architecture	Accuracy	F1-Score	Observation
SimpleRNN	0.8803	0.8794	Weak baseline
Bi-SimpleRNN	0.8737	0.8739	Lowest performance
GRU	0.9135	0.9134	Strong gated model
LSTM	0.9135	0.9134	Similar to GRU
Bi-GRU	0.9115	0.9114	Slightly lower than GRU
Bi-LSTM	**0.9171**	**0.9171**	Best recurrent model

The recurrent model comparison shows that gated models clearly performed better than SimpleRNN. GRU and LSTM both improved over SimpleRNN by more than three percentage points. This is because GRU and LSTM use gating mechanisms that help preserve important information and reduce the vanishing gradient problem.

Bi-LSTM achieved the best recurrent result. This suggests that reading the headline in both forward and backward directions helped the model capture more useful context. However, bidirectionality did not always improve performance. Bi-GRU performed slightly worse than GRU, and Bi-SimpleRNN performed worse than SimpleRNN. This means that simply adding bidirectionality does not guarantee improvement; the base architecture must also be strong enough to use the additional context effectively.

F. Training Dynamics and Confusion Matrix

The GRU and LSTM training curves show stable convergence. GRU showed smoother reduction in training and validation loss, while LSTM also achieved competitive performance.



Figure 8: GRU – Training Loss and Accuracy Curves



Figure 9: LSTM – Training Loss and Accuracy Curves

A common pattern in the training curves is that training accuracy increases faster than validation accuracy. This indicates that the deep models learn the training examples well but begin to show signs of overfitting after a few epochs. Early stopping helped prevent severe overfitting by stopping training when validation performance stopped improving.

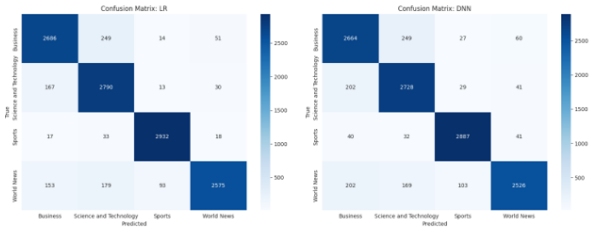


Figure 10: Confusion Matrices – LR (left) and DNN (right)

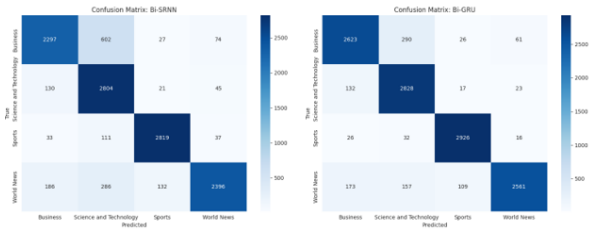


Figure 11: Confusion Matrices – Bi-SimpleRNN (left) and Bi-GRU (right)

The confusion matrix analysis shows that Sports was the easiest class to classify because it contains highly specific vocabulary. Business and World News were more difficult because both can include terms related to economy, government, policy, and international events. Science and Technology also overlapped with other classes when headlines contained general technology-related words.

G. Key Findings

The main findings are:

- TF-IDF + Logistic Regression achieved the best result with 93.28% accuracy and 93.24% F1-score.
- Optimum preprocessing performed better than both raw and extreme preprocessing.
- Extreme preprocessing reduced performance because stemming and aggressive cleaning removed useful word-level signals.
- TF-IDF-based models outperformed Skip-gram-based recurrent models.
- GRU and LSTM outperformed SimpleRNN due to their gating mechanisms.
- Bi-LSTM was the best recurrent model, but it still performed lower than Logistic Regression.
- Sports was the easiest class, while Business, Science and Technology, and World News showed more overlap.

IV. Conclusion

This project compared multiple NLP preprocessing strategies, word representation techniques, and classification models for news headline topic classification. The results show that classical TF-IDF with Logistic Regression outperformed all deep learning models, achieving 93.28% test accuracy and 93.24% macro F1-score. This indicates that sparse n-gram features are highly effective for short-text classification.

The study also shows that preprocessing strategy is very important. Optimum preprocessing improved performance by removing noise while preserving meaningful word information. In contrast, extreme preprocessing harmed performance because aggressive stemming and stopword removal reduced discriminative features. Future work may include pre-trained embeddings, transformer-based models such as BERT and RoBERTa, data augmentation, and broader hyperparameter tuning.

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